



PHYSICS CH.2 — HEAT AND THERMODYNAMICS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Joule's Law of Heating: Formula & Applications
2. Thermal Properties of Materials
3. Temperature Scales & Conversions (Conceptual + Numerical)
4. Change of State: Melting, Fusion & Latent Heat
5. Power Plants & Energy Sources
6. ⚠ Traps / Key Differences & Formula Cheat Sheet (Q363–Q380)

1. Joule's Law of Heating: Formula & Applications

- Joule's Law of Heating: Circuit mein resistance dwara electrical energy ko heat energy mein convert karne ki rate $H = I^2RT$ se di jaati hai. [Q363, Q371, Q374]
 - H = Heat produced (joules mein), I = Current flowing, R = Resistance of conductor, T = Time period of current flow. [Q374]
- Is principle par kaam karne wale devices: Immersion rod, Electric iron, Electric kettle, Electric toaster, Electric oven, Electric heater, Electric bulb, Electric fuse. [Q363, Q365, Q371, Q372]
- NOT based on Joule's heating:
 - Electric bell → magnetic effect of current par kaam karta hai, thermal effect par nahi. [Q363, Q372]
 - Electromagnet → magnetic effect par based hai. [Q363]
 - Speaker → electromagnetic induction par based hai. [Q363]
 - Electric plugs and switches → simple connectors/mechanical devices hain, heating principle par nahi. [Q365]
- Fuse bhi Joule's heating principle par kaam karta hai — jyada current flow hone par wire heat hokar melt ho jaata hai, circuit break ho jaata hai. [Q365]

2. Thermal Properties of Materials

- Tungsten (Wolfram) ka melting point sabse zyada kisi bhi metal mein — 3422°C . [Q364]
 - Isliye bulb filaments mein use hota hai — badi heat ke bawajood melt nahi hota. [Q364]
 - Tungsten exceptionally strong aur brittle hota hai. [Q364]
 - Iska thermal expansion coefficient sabse kam hai kisi bhi metal mein. [Q364]
- Specific Heat Capacity: Ek gram substance ka temperature 1°C badhane ke liye kitni heat chahiye. [Q368]
 - Water ki specific heat sabse zyada common substances mein — $4.186 \text{ J/g}^\circ\text{C}$. Isliye water climate regulate karne mein madad karta hai. [Q368]
 - Order: Water > Snow/Ice > Aluminum > Kerosene (roughly). [Q368]

3. Temperature Scales & Conversions (Conceptual + Numerical)

- Teen common temperature scales: Fahrenheit ($^\circ\text{F}$), Celsius ($^\circ\text{C}$), Kelvin (K). Kelvin = SI unit of temperature. [Q369]
- Conversion formulas (hamesha yaad rakho):
 - $\text{K} \rightarrow \text{C}$: $\text{C} = \text{K} - 273.15$ (ya approximately $\text{K} - 273$). [Q369, Q376]

- $C \rightarrow K: K = C + 273.15$ (ya approximately $C + 273$). [Q369, Q376, Q379, Q380]
- $C \rightarrow F: F = (9/5)C + 32$. [Q369, Q375, Q377, Q378]
- $F \rightarrow C: C = (F - 32) \times 5/9$. [Q369, Q375]
- Important fixed points:
 - Absolute Zero = $0\text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}$ — temperature ka theoretical minimum jahan molecular motion ruk jaati hai. [Q376]
 - Water freezing point = $0^\circ\text{C} = 273.15\text{ K} \approx 32^\circ\text{F}$. [Q369, Q378]
 - Water boiling point (sea level) = $100^\circ\text{C} = 373.15\text{ K} \approx 212^\circ\text{F}$. [Q369, Q379]
 - Normal human body temperature = $37^\circ\text{C} = 98.6^\circ\text{F} = 310.15\text{ K}$. [Q367, Q369, Q375]
 - Standard room temperature = $20^\circ\text{C} = 293\text{ K} \approx 68^\circ\text{F}$. [Q380]
- Numerical examples:
 - $37^\circ\text{C} \rightarrow ^\circ\text{F}: F = (9/5) \times 37 + 32 = 66.6 + 32 = 98.6^\circ\text{F}$. [Q375]
 - $-200^\circ\text{C} \rightarrow ^\circ\text{F}: F = (9/5) \times (-200) + 32 = -360 + 32 = -328^\circ\text{F}$. [Q377]
 - $0^\circ\text{C} \rightarrow ^\circ\text{F}: F = (9/5) \times 0 + 32 = 32^\circ\text{F}$. [Q378]
 - $100^\circ\text{C} \rightarrow \text{K}: K = 100 + 273 = 373\text{ K}$. [Q379]
 - $20^\circ\text{C} \rightarrow \text{K}: K = 20 + 273 = 293\text{ K}$. [Q380]
 - $-273.15^\circ\text{C} \rightarrow \text{K}: K = -273.15 + 273.15 = 0\text{ K}$ (Absolute Zero). [Q376]
- Inventors yaad rakho: Anders Celsius $\rightarrow$ Celsius scale; Lord Kelvin $\rightarrow$ Kelvin scale (1848 mein). [Q379]
- Ignition temperature = wo minimum temperature jis par fuel aag pakad ke jalna shuru kar de — Neutral temperature aur Boiling temperature dono alag concepts hain. [Q367]

4. Change of State: Melting, Fusion & Latent Heat

- Melting (Fusion) ke dauran temperature change nahi hota — supplied heat intermolecular bonds todne mein use hoti hai. [Q370, Q373]
- Latent Heat of Fusion = unit mass ko solid se liquid mein convert karne ke liye heat energy (bina temperature change ke). SI unit = J/kg . [Q370]
- 0°C par liquid water ke particles mein ice ke particles se zyada energy hoti hai kyunki ice ne fusion ke dauran latent heat absorb ki hai apne surroundings se.
 - Yeh absorbed energy ice crystals ke bonds todne mein gayi, jisse water molecules ki potential energy badh gayi. [Q373]
- ⚠ Trap: Temperature same (0°C) rehta hai dono states mein, lekin energy alag hai kyunki latent heat absorb/release hoti hai. [Q370, Q373]

5. Power Plants & Energy Sources

- Thermal power plant = fossil fuels (coal, oil, gas) jalakar steam banate hain jo turbines chalata hai. [Q366]
- Nuclear power plant = nuclear fission se heat generate hoti hai (atoms split hote hain reactor mein). [Q366]
- Hydro electric power plant = renewable source; dam se water flow alter karke turbines chalate hain. [Q366]
- Solar power plant = solar radiation (light + heat) ko electricity mein convert karte hain. [Q366]

6. ⚠ Traps / Key Differences & Formula Cheat Sheet (Q363–Q380)

- ⚠ Trap: Electric bell magnetic effect par kaam karta hai, Joule's heating par nahi — baar-baar galat hota hai.

[Q363, Q372]

- ⚠ Trap: Joule's heating formula $H = I^2RT$ hai — sirf I^2R ya IRt nahi. Time (T) bhi factor hai. [Q371]
- ⚠ Trap: Tungsten ka melting point 3422°C hai — 300°C ya 1000°C nahi. [Q364]
- ⚠ Trap: Melting ke dauran temperature constant rehta hai — increase ya decrease nahi hota. [Q370]
- ⚠ Trap: Kelvin se Celsius mein convert karne ke liye 273 subtract karte hain, add nahi. [Q369, Q376]
- ⚠ Trap: Water ki specific heat sabse zyada hoti hai common substances mein — yaad rakho. [Q368]
- ⚠ Trap: Ignition temperature = fuel jalne ka minimum temp; Boiling temperature = alag cheez; Neutral temperature = thermocouple ka concept. [Q367]
- ⚠ Trap: Absolute zero $0\text{ K} = -273.15^\circ\text{C}$ hai — -273 K nahi. [Q376]
- ⚠ Trap: Celsius se Fahrenheit formula mein 32 add hota hai, subtract nahi. [Q375, Q377, Q378]